THE HUMAN BATTERY™ PROJECT

YOUR TESTS, EXPLAINED

WHAT WE MEASURE AND WHY

Thirty-three markers, four functional tests, one finger prick. Read as four systems. Measured twice.

HOW TO READ THIS

Four systems, not thirty-three numbers

Every test on the list belongs to one of four systems. Together they tell us how well your cellular batteries are being charged, what is draining them, how much you can spend, and what they are built from.

CHARGE
How well fuel becomes energy. Six markers.

DRAIN
What runs in the background. Nine markers.

OUTPUT
What you can spend. Nine markers.

RESERVE
What you are built from. Nine markers.

Most lab reports hand you a wall of numbers with a flag next to the ones outside "normal." We do something different. Each marker is scored against where a well-functioning body sits, not against the population average, and the scores roll up into four systems and one Battery Score. A result at the edge of normal on a standard report can be a low score here. That is the point.

For each test you will see two things. **What it is**, in plain words. And **why we measure it**, which is the part most reports leave out.

Your Battery Score is a proxy, assembled from what can be measured today. It is not a validated clinical measure.

One thing to know before you read

A normal result on any of these does not prove every cell in your body is working perfectly. These numbers describe the environment your cells work in. They are the best indirect view we have, and measuring them twice, ninety days apart, is what makes them useful.

> **Out of range means a referral.** If any marker falls outside the laboratory's reference range, you will be referred to a physician. Every time. That is a rule, not a judgment call.

SYSTEM 1 OF 4

CHARGE

How well your body turns fuel into energy



PROTONS PUMPED ACROSS THE MEMBRANE

1

CHARGE

How well your body turns fuel into energy

01

Fasting glucose

WHAT IT IS The sugar in your blood after not eating overnight.

WHY WE MEASURE IT It is the baseline. If it is high while you are fasting, your body is having trouble putting fuel away.

02

Fasting insulin

WHAT IT IS The hormone that moves sugar out of your blood and into your cells.

WHY WE MEASURE IT This goes up years before glucose does. High insulin means your cells are ignoring the signal, and your body is shouting louder to be heard. It is the earliest warning we have.

03

HbA1c

WHAT IT IS Your average blood sugar over the last three months.

WHY WE MEASURE IT One number that shows the whole quarter, not just this morning. Ninety days is exactly one HbA1c cycle, which is why the program is ninety days.

04

HOMA-IR

WHAT IT IS A calculation from glucose and insulin together.

WHY WE MEASURE IT It tells us how hard your body is working to keep sugar normal. A high number means the fuel is in the tank but cannot get to the engine.

05

Triglycerides

WHAT IT IS Fat traveling in your blood.

WHY WE MEASURE IT When you eat more fuel than you burn, especially sugar, it turns into this. High triglycerides mean the fuel system is overloaded.

06

Triglyceride to HDL ratio

WHAT IT IS Triglycerides divided by your good cholesterol.

WHY WE MEASURE IT One of the best simple signs of whether your cells are handling fuel well. Low is good.

SYSTEM 2 OF 4

DRAIN

What is quietly using up your energy in the background

CHARGED, AND LEAKING



INFLAMMATION RUNS IN THE BACKGROUND

2

DRAIN

What is quietly using up your energy in the background

01

hs-CRP

WHAT IT IS A protein your liver makes when there is inflammation anywhere in the body.

WHY WE MEASURE IT This is the main drain marker. Inflammation is like an app running in the background on your phone. You did not open it, but it is using battery all day.

02

GGT

WHAT IT IS A liver enzyme.

WHY WE MEASURE IT It rises when the liver is stressed by alcohol, sugar, or oxidative load. It is one of the earliest signs the drain is on.

03

ALT

WHAT IT IS A liver enzyme.

WHY WE MEASURE IT Tells us whether the liver, your main fuel-processing organ, is under strain.

04

AST

WHAT IT IS A liver and muscle enzyme.

WHY WE MEASURE IT Read alongside ALT. Together they show whether the liver is keeping up.

05

Uric acid

WHAT IT IS A waste product from breaking down certain foods and from fructose.

WHY WE MEASURE IT High uric acid tracks with high sugar intake and with inflammation. It is a drain marker that moves fast when the diet changes.

06

White blood cell count

WHAT IT IS The number of immune cells in your blood.

WHY WE MEASURE IT Slightly high all the time means the immune system never gets to rest. That is a constant, quiet drain.

07

Homocysteine

WHAT IT IS An amino acid that builds up when your B vitamins are low or not working.

WHY WE MEASURE IT High homocysteine is hard on blood vessels and points to a methylation problem, which affects how your cells repair.

08

AA to EPA ratio

WHAT IT IS The balance between an inflammatory fat and an anti-inflammatory fat in your cells.

WHY WE MEASURE IT This is the fat side of the drain. It comes from what you eat, and it moves when you eat sardines instead of seed oils.

09

Kynurenine to tryptophan ratio

WHAT IT IS How much of an important amino acid is being burned up by inflammation.

WHY WE MEASURE IT When inflammation is high, your body diverts tryptophan away from where it should go. This ratio shows the diversion. It also connects to some of the newest research in the model.

SYSTEM 3 OF 4

OUTPUT

How much power you can actually spend

STORED, THEN SPENT



HORMONES DECIDE HOW MUCH YOU CAN USE

3

OUTPUT

How much power you can actually spend

01 **TSH**
WHAT IT IS The signal your brain sends to your thyroid.
WHY WE MEASURE IT The thyroid sets the speed of every cell. TSH tells us if the brain is having to shout to get it going.

02 **Free T3**
WHAT IT IS The active thyroid hormone.
WHY WE MEASURE IT This is the one your cells actually use. Low free T3 feels like cold hands, slow thinking, and no drive, even when TSH looks fine.

03 **Free T4**
WHAT IT IS The storage form of thyroid hormone.
WHY WE MEASURE IT Your body converts T4 to T3. Reading both tells us if the conversion is working.

04 **Total testosterone**
WHAT IT IS The main drive and repair hormone, in both men and women.
WHY WE MEASURE IT Low testosterone feels like no motivation, slow recovery, and no strength gains. Scored against your own sex's range.

05 **Free testosterone**
WHAT IT IS The testosterone that is actually available to your cells.
WHY WE MEASURE IT Total can look fine while free is low. This is the number that matches how you feel.

06 **DHEA-S**
WHAT IT IS A hormone from the adrenal glands that your body makes other hormones from.
WHY WE MEASURE IT It is a reserve measure for the whole hormone system. It drops with age and with chronic stress.

07 **Morning cortisol**
WHAT IT IS Your main stress hormone, measured when it should be highest.
WHY WE MEASURE IT Cortisol should be high in the morning and fall through the day. Too low in the morning means the system is worn down. Drawn at the same hour both times because it changes across the morning.

08 **IGF-1**
WHAT IT IS A growth and repair signal.
WHY WE MEASURE IT Tells us how well your body is rebuilding. Too low means slow repair. Too high is not the goal either.

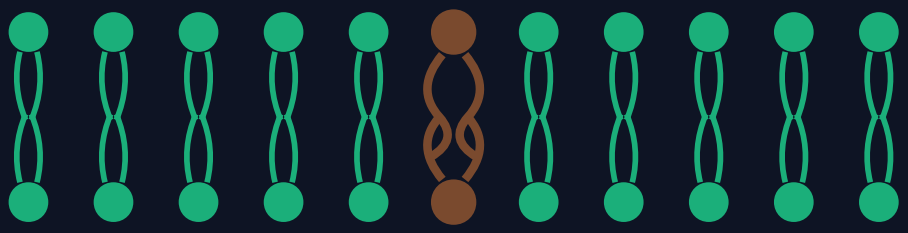
09 **SHBG**
WHAT IT IS A protein that binds hormones and controls how much is free.
WHY WE MEASURE IT It explains the gap between total and free testosterone, and it moves with insulin, so it connects Output back to Charge.

SYSTEM 4 OF 4

RESERVE

What your batteries are built from

THE MEMBRANE, UP CLOSE



DHA IS THE FLEXIBLE ONE

WHAT THE BATTERY IS BUILT FROM

4

RESERVE

What your batteries are built from

01

Omega-3 Index

WHAT IT IS The percentage of your red blood cell membranes made of EPA and DHA.

WHY WE MEASURE IT Required at day 0 and day 90. DHA is what the membranes that receive the light signal are built from, the retina most of all. Low DHA means the signal lands on a degraded receiver. This is the most responsive marker on the panel and the most direct check on whether the protocol is working. Done at home with a finger prick, not at the lab.

02

25-OH vitamin D

WHAT IT IS The storage form of vitamin D.

WHY WE MEASURE IT The direct readout on the sun protocol. Almost everyone starts low. It affects immune function, hormones, and mood.

03

Ferritin

WHAT IT IS Your stored iron.

WHY WE MEASURE IT Too low means the batteries cannot carry oxygen well. Too high can mean inflammation. Read together with iron saturation.

04

Iron saturation

WHAT IT IS How much of your iron-carrying protein is actually carrying iron.

WHY WE MEASURE IT Ferritin tells us the stockpile. Saturation tells us what is in use. You need both to know the real story.

05

Vitamin B12

WHAT IT IS A vitamin your nerves and red blood cells depend on.

WHY WE MEASURE IT Low B12 feels like fatigue and fog and tingling. It is common in people who do not eat much animal protein.

06

Folate

WHAT IT IS A B vitamin that works with B12 for repair and methylation.

WHY WE MEASURE IT Read with B12 and homocysteine. The three together show whether the repair system has what it needs.

07

RBC magnesium

WHAT IT IS Magnesium inside your red blood cells, not just in the blood around them.

WHY WE MEASURE IT Regular serum magnesium is almost always normal, even when you are low. This is the honest version. Magnesium runs hundreds of the reactions that make and use energy.

08

Albumin

WHAT IT IS The main protein in your blood.

WHY WE MEASURE IT A simple measure of whether you have enough protein and whether your liver is making it. Low albumin means the raw material is short.

09

HDL

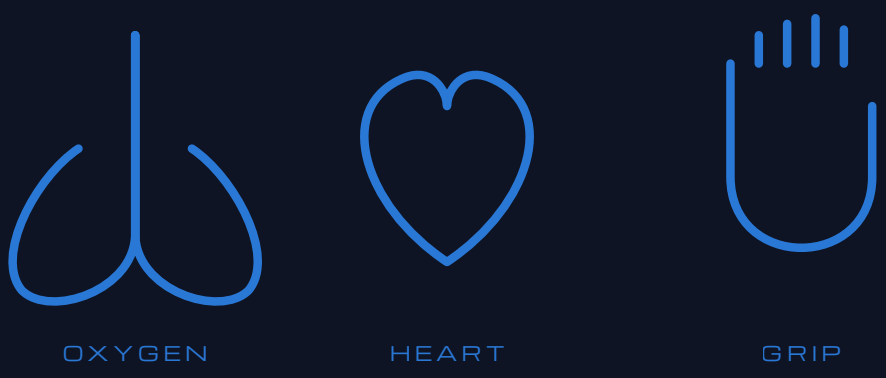
WHAT IT IS The good cholesterol.

WHY WE MEASURE IT It is protective, and it is in Reserve because it is a marker of what your body has to work with, not a drain.

FUNCTIONAL TESTS

What blood cannot show

Oxygen moved, force produced, and how fast the system recovers. Four measurements you take yourself.



FUNCTIONAL TESTS

The four things blood cannot show

01

VO_2max

WHAT IT IS How much oxygen your body can take in and use when you are working as hard as you can.

WHY WE MEASURE IT This is the Charge test that blood cannot show. Oxygen is the last step of the reaction that builds the voltage in every cell. VO_2max is the ceiling on that reaction for your whole body, and it is the strongest predictor of how long people live that exists. A lab test with a mask is best. A step test at home works if you do it the same way both times.

02

Grip strength

WHAT IT IS How hard you can squeeze.

WHY WE MEASURE IT It is a stand-in for total body strength and it predicts health outcomes surprisingly well. Measured with a small device you squeeze three times. Takes one minute.

03

Resting heart rate

WHAT IT IS How fast your heart beats when you are completely at rest.

WHY WE MEASURE IT Lower usually means a stronger, more efficient system. Take it first thing in the morning, before you get up.

04

Blood pressure

WHAT IT IS The force of blood against your artery walls.

WHY WE MEASURE IT Read at the same time of day. It responds to sun, sleep, minerals and cold faster than almost anything else on this list.

GETTING IT DONE

Where to go

One walk-in draw. One kit in the mail. Same lab both times.



WHERE TO GET IT DONE

Getting your blood drawn

What to look for in a draw site

Your Human Battery Score is identical at every panel. Higher panels add context, not points.

The specification, and any provider meeting it will do: walk-in or same-week appointments, no physician order required, a custom panel built from a marker list you bring, and samples sent to a national reference laboratory. Several national chains and online-order services meet all four, and most cities have at least one.

Some markers may not be on a standard menu. RBC magnesium, the AA to EPA ratio and the kynurenine to tryptophan ratio are the ones most likely to need a special request. Ask. If a marker genuinely cannot be run, the program excludes it from your score rather than guessing, so you are not penalized.

Use the same one twice

Whichever provider you choose, use the same one at day 90. Reference ranges and assay methods differ between laboratories, and a change of provider between your two draws puts a difference in the numbers that did not happen in you.

The Omega-3 Index

Not done at the lab. Order a dried-blood-spot Omega-3 Index test: the specification is a lab reporting EPA plus DHA as a percentage of red blood cell fatty acids. OmegaQuant originated that measurement and sells one; we take no commission on it. It arrives in the mail. You prick your finger, put a drop on the card, and mail it back. Allow two to four weeks from ordering to result. Do it the same week as your blood draw, both times.

Order two at once. You need one for day 0 and one for day 90, sellers usually discount the pair, and the second kit is then already in your drawer when day 90 comes.

Order the version that also reports the AA to EPA ratio. The cheaper option reports the Omega-3 Index on its own. The AA to EPA ratio is on your panel and is the marker most likely to be missing from a standard lab menu, so the cheaper option costs you a marker.

VO_2max and grip

A VO_2max lab test with a mask is available at most sports performance centers and university laboratories, at a cost in the low hundreds. If that is not practical, your First Steps document has a step test you can do at home. Grip strength uses a small hand dynamometer, inexpensive online, that you keep for day 90.

BEFORE THE DRAW

Keep the numbers honest

Day 0 and day 90 only mean something if they were taken the same way.

SAME WAY, BOTH TIMES



7 TO 9 AM

12 HOURS EMPTY

6

BEFORE THE DRAW

Six rules

1 **Fast for 12 hours**
Nothing but water from the night before. Coffee counts as food for this. The morning glass with lemon and salt is fine.

2 **Same time of morning, both draws**
Between 7 and 9am. Cortisol and testosterone both change across the morning, and a 7am draw compared to an 11am draw shows a change that is not real.

3 **No alcohol for 48 hours**
It moves the liver enzymes and triglycerides.

4 **No hard training for 24 hours**
It moves AST, CRP, and cortisol.

5 **Drink your water**
Dehydration makes several markers read high. Have your morning glass and a full bottle before the draw.

6 **Same lab, both times**
Different labs use different machines and reference ranges. Day 0 and day 90 must come from the same place.

> **Bring this list to the lab.** Ask for exactly these.

CHARGE fasting glucose, fasting insulin, HbA1c, triglycerides, HDL, standard lipid panel
DRAIN hs-CRP, GGT, ALT, AST, uric acid, CBC with differential, homocysteine, omega-3 fatty acid profile with AA and EPA, kynurenine and tryptophan
OUTPUT TSH, free T3, free T4, total testosterone, free testosterone, DHEA-S, AM cortisol, IGF-1, SHBG
RESERVE 25-OH vitamin D, ferritin, iron and TIBC, vitamin B12, folate, RBC magnesium, albumin
HOMA-IR and the ratios are calculated, not drawn.

On 25-OH vitamin D. Vitamin D comes from sunlight first and from cold-water fish second (sardines, herring, mackerel, salmon). If a winter blood test shows the level has fallen and sun is not available, D3 may be needed to bring it back up. That is the last step, not the first.

> The Human Battery Project is an educational wellness program, not medical treatment. It does not diagnose or treat any condition. Any result outside the laboratory's reference range is referred to a physician.